

Literature Review: Quantized Domain Specialists for Multi-Agent Scientific Reasoning

Razeen Wasif

Independent researcher

razeen.wasif66@gmail.com

June 2026

Abstract

This review situates the proposed research project (companion *Research Proposal*) at the intersection of four active research areas: multi-agent LLM systems, LLM quantization, domain-specific fine-tuning, and verification/self-checking. Each area has substantial literature individually; the intersection is under-studied. We survey ~ 55 sources to make that gap visible and to provide the grounding for the proposal’s methodology choices.

Contents

1	Introduction and Scope	2
2	Multi-Agent LLM Systems	2
2.1	Foundational systems	2
2.2	Debate as a quality mechanism	3
2.3	Self-improvement loops	3
2.4	Co-Scientist — the closest prior art	3
2.5	Reasoning primitives	3
2.6	Verification of generated content	4
3	LLM Quantization	4
3.1	Post-training quantization	4
3.2	Extreme low-bit quantization	4
3.3	Rotation-based quantization	4
3.4	Quantization-aware training	4
3.5	Quantization $\times$ reasoning — the most directly relevant prior work	5
3.6	The serving stack — llama.cpp / GGUF	5
4	Parameter-Efficient Fine-Tuning	5
4.1	LoRA and family	5
4.2	Domain expansion without forgetting	5
4.3	Model merging and sparse upcycling	6
5	Domain-Specific Scientific LLMs	6
5.1	Encoder-era scientific models	6
5.2	Generative scientific LLMs	6
5.3	Continued pretraining methodology	6
6	Tool Use, MCP, and Retrieval	7

6.1	Tool-augmented LLMs	7
6.2	Model Context Protocol	7
6.3	Retrieval-Augmented Generation	7
6.4	Tool-augmented scientific reasoning	7
7	AI for Scientific Discovery	7
7.1	Landmark systems	7
7.2	Surveys of LLMs in science	7
8	Open-Weights Base Models	8
9	Evaluation Benchmarks	8
9.1	General reasoning	8
9.2	LLM-as-Judge evaluation	8
10	Information-Theoretic Perspectives	8
11	Synthesis — Where the Gap Is	9
12	References	10

1 Introduction and Scope

The proposed research project sits at the intersection of four active research areas:

1. **Multi-agent LLM systems** — increasingly studied for their ability to outperform single-model approaches on reasoning, factuality, and scientific tasks.
2. **LLM quantization** — a mature engineering subfield with active research on extreme compression (≤ 4 bits).
3. **Domain-specific fine-tuning** — a long-standing tradition (SciBERT, BioBERT) recently extended into the generative-LLM era (Galactica, Med-PaLM, PMC-LLaMA).
4. **Verification and self-checking** — emerging methods for catching LLM errors via debate, self-critique, process supervision, and external tool use.

Each area has substantial literature individually. **The intersection is under-studied.** No prior work that we have found combines: (a) a heterogeneous multi-agent system, (b) with a quantized domain-specialist as a *verification* component, (c) subjected to a *controlled quantization ablation*. This review is organized to make that gap visible.

2 Multi-Agent LLM Systems

2.1 Foundational systems

The multi-agent paradigm for LLMs took shape in 2023. **AutoGen** [1] introduced a conversational multi-agent framework where customizable, conversable agents (LLM-driven, human-driven, or tool-driven) cooperate to solve problems. The paper demonstrated effectiveness across mathematics, coding, QA, and operations research. **MetaGPT** [2] extended this with Standard Operating Procedure (SOP) encoding: agents are assigned roles (project manager, architect, developer) that mirror real software-team workflows, producing more coherent end-to-end software artifacts. **CAMEL** [3] explored autonomous role-playing between AI agents (assistant + user) to generate task-oriented conversations, producing the 25K-conversation AI Society dataset as a side artifact.

A complementary line — **HuggingGPT** [4] — used an LLM as a *controller* orchestrating dozens of specialized HuggingFace models. The four-stage workflow (task planning → model selection → execution → response generation) is conceptually related to our verification layer, where a frontier model dispatches claim-checking to local specialists.

2.2 Debate as a quality mechanism

Du et al. [5] is the most cited theoretical result in this space. They show that multiple LLM instances proposing and critiquing each other’s answers over multiple rounds consistently outperform single-model baselines on mathematics, strategic reasoning, and factual QA. **Arithmetic accuracy improves with both more agents and more debate rounds** — a scaling law that motivates the multi-voice design of Council and /discover.

2.3 Self-improvement loops

Three related approaches established that LLMs can improve their own outputs without weight updates:

- **Self-Refine** [6] — a single LLM generates, critiques, and refines its output iteratively. Reports ~20% absolute improvement across diverse tasks vs. direct generation.
- **Reflexion** [7] — agents verbally reflect on task feedback and store reflections in an episodic memory buffer to improve subsequent attempts.
- **Tree of Thoughts** [8] — generalizes Chain-of-Thought to deliberate search over reasoning branches, with self-evaluation and backtracking. Lifts GPT-4’s Game-of-24 success rate from 4% (CoT) to 74%.

All three operate within a *single model*. The present project extends this lineage by introducing a *different* model (a specialist) as the critic, betting that cross-model heterogeneity catches errors a single model is blind to.

2.4 Co-Scientist — the closest prior art

Gottweis et al. [9] introduce Google’s Gemini-2.0-powered multi-agent system with six specialized agents (Generation, Reflection, Ranking, Proximity, Evolution, Meta-review) operating in a closed tournament loop over a hypothesis pool. **This is the closest published architecture to the planned Co-Scientist mode of our system.** Critical differences in the present work:

- All Google Co-Scientist agents are frontier (Gemini); we mix frontier reasoners + local specialists.
- Google Co-Scientist is closed-source; our verification-layer hybrid is reproducible on consumer hardware.
- Google does not study the *quantization* of any component; that is the load-bearing contribution of our proposal.

2.5 Reasoning primitives

Chain-of-Thought prompting [10] established that intermediate reasoning steps substantially improve LLM performance on math, commonsense, and symbolic tasks — but only emerges around ~100B parameters. Tree of Thoughts [8] generalizes this. **Process Reward Models** [11] — OpenAI’s “Let’s Verify Step by Step” work — show that *step-level* supervision substantially outperforms outcome-only supervision on math benchmarks. The verifier architecture in the present project draws on PRM principles: it audits *individual claims*, not just final answers.

2.6 Verification of generated content

SelfCheckGPT [12] detects hallucinations in black-box LLMs by sampling multiple responses and measuring consistency — a method that scales to closed APIs without internal access. **Constitutional AI** [13] introduced RLAIFF: training an AI to evaluate outputs against a constitution of natural-language principles, replacing human feedback. **LLM-as-a-Judge** [14] systematically validated using strong LLMs (GPT-4) as judges of chatbot output quality, finding $\sim 80\%$ agreement with human raters while characterizing position/verbosity/self-enhancement biases.

These verification techniques all use a *strong* LLM as judge. The present project’s open question: can a small, quantized, fine-tuned model serve as a credible domain-specific judge despite its lower general capability?

3 LLM Quantization

3.1 Post-training quantization

GPTQ [15] introduced layer-wise weight quantization for LLMs using inverse-Hessian information to update remaining weights as each one is quantized. Established that 3–4 bit weight-only quantization could be done without catastrophic accuracy loss on large models. **AWQ** [16] refined this with activation-aware per-channel scaling, observing that protecting $\sim 1\%$ of salient weights preserves most of the model’s accuracy. AWQ has been adopted by NVIDIA TensorRT-LLM, AMD, Google Vertex, Amazon SageMaker, vLLM, and HuggingFace TGI.

SmoothQuant [17] addressed the *activation* quantization problem: activations have outliers $\sim 100\times$ larger than typical values, making W8A8 (8-bit weight + activation) quantization lose accuracy. SmoothQuant migrates the quantization burden from activations to weights via a mathematically-equivalent transformation. **ZeroQuant** [18] introduced fine-grained, hardware-friendly quantization with layer-by-layer knowledge distillation, achieving INT8 W/A quantization with minimal accuracy impact and up to $5\times$ speedup.

3.2 Extreme low-bit quantization

OmniQuant [19] combined learnable weight clipping with sample-efficient calibration to push compression to 2–4 bits. **BitNet b1.58** [20] — Microsoft’s 1.58-bit model — trains directly in low-bit precision from scratch using ternary weights $\{-1, 0, 1\}$, encoding $\log_2(3) \approx 1.58$ bits per parameter. Matches FP16 LLaMA in perplexity at 3B parameter scale while using $3.55\times$ less GPU memory and running $2.71\times$ faster. The most extreme PTQ work — AQLM [21], QuIP, OneBit — pushes below 3 bits with diminishing accuracy returns. *Below 3 bits remains research territory; production deployments stay at Q4–Q8.*

3.3 Rotation-based quantization

A 2024 advance: QuaRot showed that rotating LLM hidden states (via Hadamard transforms) removes outliers without changing the model’s output, dramatically easing quantization. **SpinQuant** [22] extended this by *learning* the rotation matrices via Cayley optimization on a small calibration set. For Llama-2 7B at W4A4KV4, SpinQuant narrows the accuracy gap to FP16 to 2.9 points — substantially better than LLM-QAT [23] (+19.1 pts gap) and SmoothQuant (+25 pts).

3.4 Quantization-aware training

LLM-QAT [23] introduced data-free QAT for LLMs: generates synthetic training data from the pretrained model itself (no need for the original training data), distills the FP16 model into a quantized student. Quantizes weights, activations, *and* KV cache. Demonstrated on LLaMA

7B/13B/30B down to 4 bits. **The present project will use LLM-QAT-style methodology for Phase 3’s QAT vs. PTQ comparison.**

3.5 Quantization × reasoning — the most directly relevant prior work

Liu et al. [24] is the closest prior work to our central question. They evaluate quantization on math reasoning specifically, finding:

- AWQ and GPTQ introduce up to **32.39% accuracy degradation** (average 11.31%) on Llama-3 models, concentrated in numerical computation and reasoning planning.
- Performance drops on math reasoning are disproportionately larger than on general common-sense and language-understanding benchmarks.
- **Fine-tuning quantized models on only 545 task-specific examples for 3 minutes on 4 GPUs effectively restores reasoning capabilities to near full-precision levels.**

This last finding is directly relevant: it suggests that small-scale specialty fine-tuning may compensate for quantization damage, which is exactly the verification-layer hypothesis. The present project extends this by (a) using the fine-tuned quantized model as a *verifier* rather than a primary reasoner, (b) measuring across more bit-depths with explicit PTQ-vs-QAT comparison, (c) embedding the result in a multi-agent system.

3.6 The serving stack — llama.cpp / GGUF

The community-standard format for serving quantized LLMs is GGUF (introduced in `llama.cpp` [25]). GGUF supports 1.5–8 bit quantization via “K-quant” variants (Q2_K, Q3_K_S/M/L, Q4_K_S/M, Q5_K_S/M, Q6_K) that use mixed-precision blocks for better accuracy at the same effective bit budget. The present project’s Phase 3 quantization sweep uses GGUF formats because they are widely-deployed, well-tested, and serve directly through Ollama.

4 Parameter-Efficient Fine-Tuning

4.1 LoRA and family

LoRA [26] is the foundational method: freeze pre-trained weights, introduce trainable low-rank decomposition matrices in parallel. Achieves up to 10,000× fewer trainable parameters and 3× less GPU memory for GPT-3-scale fine-tuning, with no inference-time latency cost (adapters can be merged).

QLoRA [27] combined LoRA with 4-bit quantization of the frozen base, enabling fine-tuning of 65B models on a single 48GB GPU. Introduced 4-bit NormalFloat (NF4) quantization and Double Quantization. *Demonstrated that combining quantization + adapter fine-tuning is viable.* The present project inherits this lineage but flips the direction: instead of QLoRA’s “fine-tune through frozen quantization,” we explore “fine-tune at FP16, then quantize the merged result, and measure the curve.”

DoRA [28] decomposes pretrained weights into *magnitude* and *direction* components, applying LoRA only to direction. Consistently outperforms LoRA across LLaMA, LLaVA, and VL-BART on commonsense reasoning, visual instruction tuning, and image/video-text tasks. *No additional inference cost.*

4.2 Domain expansion without forgetting

LLaMA-Pro [29] proposes *block expansion*: extending a pretrained LLM with zero-initialized Transformer blocks tuned exclusively on the new corpus while original blocks remain frozen. Specifically addresses catastrophic forgetting in domain specialization. LLaMA Pro-8.3B (from

LLaMA-2-7B) matches StarCoder-15B on code and beats LLaMA-2-13B and LLaMA-2-34B on math while remaining smaller. Directly relevant: it is a method for adding domain expertise *without losing general capability* — a key requirement for the verifier role.

4.3 Model merging and sparse upcycling

Task Arithmetic introduced the concept of “task vectors” — the difference between fine-tuned and pre-trained weights — that can be added, subtracted, or interpolated. TIES-Merging addresses interference between merged models by resolving sign conflicts. DARE randomly drops up to 90% of delta parameters before merging, mitigating interference. **Sparse Upcycling** [30] trains MoE models from dense checkpoints by replacing MLP layers with MoE layers initialized from the original MLP, recovering $\sim 100\%$ of the dense-training cost while reaching MoE-quality outputs at $\sim 50\%$ incremental cost.

5 Domain-Specific Scientific LLMs

5.1 Encoder-era scientific models

SciBERT [31] was the first widely-used pretrained encoder for scientific text — BERT pretrained on 1.14M biomedical + CS papers from Semantic Scholar with an in-domain vocabulary (SciVocab). **BioBERT** narrowed the focus to PubMed/PMC. These models established the *value of domain pretraining* but were limited to discriminative tasks. **ChemBERTa** [32] extended this to chemistry, pretrained on SMILES strings via masked language modeling.

5.2 Generative scientific LLMs

Galactica [33] — Meta’s 120B parameter LLM trained on 106B tokens of curated scientific text. Outperformed Chinchilla on math MMLU (41.3% vs 35.7%), beat PaLM 540B on MATH (20.4% vs 8.8%). *Famously, the demo was pulled within 3 days after public hallucination examples.* The technical contribution stands; the deployment lesson — that even strong domain models hallucinate confidently — is one of the motivations for the verification-layer approach.

Med-PaLM 2 [34] — Google’s medical specialist built from PaLM 2 + medical domain fine-tuning + new prompting strategies. Achieved 86.5% on MedQA. **PMC-LLaMA** [35] — open-source biomedical LLM, two-stage training: pretraining on 4.8M biomedical papers + 30K medical textbooks, then instruction tuning. Outperformed ChatGPT on PubMedQA, MedMCQA, USMLE. *Demonstrates the open-source path to specialist models that the present project follows.*

5.3 Continued pretraining methodology

Gururangan et al. [36] established **DAPT** (Domain-Adaptive Pretraining) and **TAPT** (Task-Adaptive Pretraining) as canonical methodology: take a general pretrained model, continue masked-language-modeling on a target-domain corpus, then fine-tune on the end task. This is the methodology the present project’s Phase 2 follows.

Catastrophic forgetting [37] — an empirical study finding that catastrophic forgetting in LLMs *intensifies with scale* (worse in 7B than 1B). Models suffer stronger forgetting in domain knowledge, reasoning, and reading comprehension. Critical risk to address in Phase 2; mitigations include general instruction mixing during fine-tuning and methods like LLaMA-Pro’s block expansion.

6 Tool Use, MCP, and Retrieval

6.1 Tool-augmented LLMs

Toolformer [38] — LLMs learn to use external tools (calculator, search, translation, QA, calendar) by self-supervised generation of training examples with tool API calls. **ReAct** [39] — interleaves reasoning traces with tool actions, demonstrating that explicit Reason+Act traces beat pure-reasoning or pure-action baselines. ReAct’s structured tool-use format has become the basis for most modern function-calling implementations from OpenAI, Anthropic, and Google.

6.2 Model Context Protocol

MCP [40] — open standard for connecting LLM applications to external data sources and tools. Released November 2024. Adopted by OpenAI (March 2025) and Google. Solves the “ $M \times N$ problem” of $LLM \times tool$ integration via a standard JSON-RPC client-server protocol. Three server primitives: Prompts, Resources, Tools.

The present project uses MCPs for arXiv search and Wolfram Alpha computation. The MCP layer is the third tier in the system architecture (after frontier reasoners and local specialists).

6.3 Retrieval-Augmented Generation

RAG [41] — the seminal “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” Introduced the parametric + non-parametric memory split: a pretrained seq2seq model (BART) for parametric knowledge, an external corpus (Wikipedia, indexed via Dense Passage Retrieval) for non-parametric. Set state-of-the-art on three open-domain QA benchmarks. The progenitor of an enormous downstream ecosystem.

6.4 Tool-augmented scientific reasoning

SciAgent [42] — tool-augmented LLMs for scientific reasoning, introducing the SciToolBench benchmark. Explicitly tests whether providing scientific tools improves LLM performance on domain-specific questions. Directly relevant to the MCP-augmentation Phase 4 of the present project.

7 AI for Scientific Discovery

7.1 Landmark systems

AlphaFold 3 [43] — joint Google DeepMind + Isomorphic Labs work; diffusion-based architecture predicting protein structures, protein-DNA/RNA complexes, post-translational modifications, ligands, and ions in a single unified model. *Order-of-magnitude accuracy improvements over specialized prior tools.*

AlphaGeometry [44] — neuro-symbolic geometry theorem prover trained from scratch on millions of synthetic theorems. Solves Olympiad-level geometry near gold-medalist standard. AlphaProof + AlphaGeometry 2 solved 4 of 6 IMO 2024 problems (silver-medal score).

These works establish that narrow, well-engineered scientific AI systems can produce frontier-of-knowledge results. They illustrate a model the present project aspires to: domain-specialized systems beat generalist models on their home turf.

7.2 Surveys of LLMs in science

Zhang et al. [45] catalog 260+ scientific LLMs across modalities (text, molecules, proteins), summarizing pretraining datasets, evaluation tasks, and gaps. The LLM4SR survey [46] covers

LLM use across hypothesis discovery, experiment planning, writing, and review. **80.9% of 800+ surveyed published authors had used LLMs in their research workflow.**

8 Open-Weights Base Models

The current generation of open-weights base models forms the candidate pool for the project’s specialist base:

- **Llama 3** [47] — Meta’s herd from 8B to 405B. The 405B model achieves GPT-4-class quality.
- **Gemma 3** [48] — Google’s open-weights family with 1B, 4B, 12B, 27B variants. Multimodal. 128K context. *The present project’s working assumption is “Gemma 4 31B”; falls back to Gemma 3 27B with no methodology change.*
- **Qwen 2.5** [49] — Alibaba’s open-weights family, especially strong at math reasoning. *A leading candidate base model for any math-specialized future variant.*
- **Phi-3** [50] — Microsoft’s small-LLM family. Demonstrates that *data curation > parameter count*.
- **Mixtral 8x7B** [51] — Mistral’s sparse Mixture-of-Experts. Apache 2.0 license.
- **DeepSeek-V3** [52] — 671B MoE with 37B active per token. Achieves 88.5 MMLU, 75.9 MMLU-Pro, 59.1 GPQA. The strongest current open-source reasoner.

For a 31B-class domain specialist, the candidate pool is: Gemma 3 27B, Gemma 4 31B (if it exists), Qwen 2.5 32B, Llama 3.x in the 30B range. Gemma is the user’s existing choice (already on disk via Ollama). Phase 2 should benchmark at least Gemma vs. Qwen on the eval before committing.

9 Evaluation Benchmarks

9.1 General reasoning

GPQA [53] — Graduate-level “Google-proof” QA: 448 multiple-choice questions in biology, physics, chemistry written by domain PhDs. Domain experts score 65%; non-expert validators score 34% even with unrestricted web access. *The current standard benchmark for scientific reasoning capability in frontier LLMs.*

MMLU-Pro [54] — extends MMLU with 10-option questions, removes trivial items, and adds reasoning-focused replacements. 12K+ questions across 14 domains including Biology, Chemistry, Computer Science, Math, Physics. Accuracy drops 16–33% vs. MMLU, with much greater robustness to prompt variation.

The present project’s eval set will include MMLU-Pro physics + chemistry subsets as the out-of-domain reference, alongside the hand-curated GW-specific in-domain eval.

9.2 LLM-as-Judge evaluation

Beyond benchmarks, LLM-as-Judge protocols [14] are increasingly used for scalable evaluation of subjective qualities. The present project uses LLM-as-Judge sparingly, preferring hand-graded rubrics for the small- N eval set in Phase 0.

10 Information-Theoretic Perspectives

A smaller but important literature frames neural-network quantization in classical rate-distortion terms [55]. The objective is to minimize representation error (distortion) for a given codebook

size (rate). Recent work establishes Gibbs-type variational formulations of rate-distortion that unify compression, quantization, and decoding as convex projections of continuous information onto discrete manifolds.

Why this matters for the present project: the author’s background research (GW SNR quantization) operates in rate-distortion territory. Connecting LLM-weight quantization to the same theoretical framework is a natural extension and is one of the project’s potential theoretical contributions — though the empirical/system contribution is the primary aim.

11 Synthesis — Where the Gap Is

Mapping the literature against the present project’s components:

Component	Covered?	Gap
Multi-agent LLM debate	Yes	None major
LLM quantization at extreme bit-depths	Yes	Mostly studies single-model, single-task degradation
Domain fine-tuning	Yes	Limited interaction with quantization
LLM-as-Judge / verification	Yes	Always uses <i>strong</i> judges; never small quantized specialists
Quantization $\times$ reasoning	Emerging	One paper [24]; doesn’t study multi-agent verification
Quantized specialist in multi-agent system	No known prior work	The project’s headline contribution

Table 1: Coverage map: where the literature is dense, and where the gap is.

The intellectual gap is specifically:

Does aggressive quantization of a domain-specialist fine-tune preserve enough capability for that specialist to act as an effective verifier in a heterogeneous multi-agent system — and where on the bit-depth curve does that capability collapse?

Each of the four ingredients (multi-agent, quantization, specialization, verification) is well-studied. **None of the published work combines all four**, which is what makes the project a coherent, citable, single-paper contribution rather than a re-implementation of an existing study.

Adjacent open questions

Out of scope for the primary contribution, but worth flagging as candidates for follow-up work:

1. Are quantization failures domain-correlated? (Liu et al. [24] suggests yes, but only one domain studied.)
2. Does QAT recover the entire gap to FP16, or only part? (LLM-QAT shows recovery; not directly tested against multi-agent eval.)
3. Does merging multiple specialists (TIES / DARE) preserve per-domain accuracy?
4. Is verification-layer effectiveness a function of voice diversity?

These are follow-up paper candidates, not primary contributions.

12 References

- [1] Q. Wu, G. Bansal, J. Zhang, Y. Wu, S. Zhang, E. Zhu, B. Li, L. Jiang, X. Zhang, and C. Wang. “AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation”. In: *Conference on Language Modeling (COLM)*. 2024. arXiv: [2308.08155](https://arxiv.org/abs/2308.08155). URL: <https://arxiv.org/abs/2308.08155>.
- [2] S. Hong et al. *MetaGPT: Meta Programming for a Multi-Agent Collaborative Framework*. 2023. arXiv: [2308.00352](https://arxiv.org/abs/2308.00352). URL: <https://arxiv.org/abs/2308.00352>.
- [3] G. Li, H. A. A. K. Hammoud, H. Itani, D. Khizbullin, and B. Ghanem. “CAMEL: Communicative Agents for “Mind” Exploration of Large Language Model Society”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2303.17760](https://arxiv.org/abs/2303.17760). URL: <https://arxiv.org/abs/2303.17760>.
- [4] Y. Shen, K. Song, X. Tan, D. Li, W. Lu, and Y. Zhuang. “HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2303.17580](https://arxiv.org/abs/2303.17580). URL: <https://arxiv.org/abs/2303.17580>.
- [5] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, and I. Mordatch. “Improving Factuality and Reasoning in Language Models through Multiagent Debate”. In: *Proceedings of the 41st International Conference on Machine Learning*. 2024. arXiv: [2305.14325](https://arxiv.org/abs/2305.14325). URL: <https://arxiv.org/abs/2305.14325>.
- [6] A. Madaan et al. “Self-Refine: Iterative Refinement with Self-Feedback”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2303.17651](https://arxiv.org/abs/2303.17651). URL: <https://arxiv.org/abs/2303.17651>.
- [7] N. Shinn, F. Cassano, E. Berman, A. Gopinath, K. Narasimhan, and S. Yao. “Reflexion: Language Agents with Verbal Reinforcement Learning”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2303.11366](https://arxiv.org/abs/2303.11366). URL: <https://arxiv.org/abs/2303.11366>.
- [8] S. Yao, D. Yu, J. Zhao, I. Shafran, T. L. Griffiths, Y. Cao, and K. Narasimhan. “Tree of Thoughts: Deliberate Problem Solving with Large Language Models”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2305.10601](https://arxiv.org/abs/2305.10601). URL: <https://arxiv.org/abs/2305.10601>.
- [9] J. Gottweis et al. *Towards an AI Co-Scientist*. Forthcoming in *Nature* (2026). 2025. arXiv: [2502.18864](https://arxiv.org/abs/2502.18864). URL: <https://arxiv.org/abs/2502.18864>.
- [10] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, and D. Zhou. “Chain-of-Thought Prompting Elicits Reasoning in Large Language Models”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2022. arXiv: [2201.11903](https://arxiv.org/abs/2201.11903). URL: <https://arxiv.org/abs/2201.11903>.
- [11] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, and K. Cobbe. *Let’s Verify Step by Step*. 2023. arXiv: [2305.20050](https://arxiv.org/abs/2305.20050). URL: <https://arxiv.org/abs/2305.20050>.
- [12] P. Manakul, A. Liusie, and M. J. F. Gales. “SelfCheckGPT: Zero-Resource Black-Box Hallucination Detection for Generative Large Language Models”. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2023. arXiv: [2303.08896](https://arxiv.org/abs/2303.08896). URL: <https://arxiv.org/abs/2303.08896>.
- [13] Y. Bai, S. Kadavath, S. Kundu, A. Askell, J. Kernion, A. Jones, A. Chen, A. Goldie, A. Mirhoseini, C. McKinnon, et al. *Constitutional AI: Harmlessness from AI Feedback*. 2022. arXiv: [2212.08073](https://arxiv.org/abs/2212.08073). URL: <https://arxiv.org/abs/2212.08073>.
- [14] L. Zheng et al. “Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2306.05685](https://arxiv.org/abs/2306.05685). URL: <https://arxiv.org/abs/2306.05685>.
- [15] E. Frantar, S. Ashkboos, T. Hoefer, and D. Alistarh. “GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers”. In: *International Conference on*

- Learning Representations (ICLR)*. 2023. arXiv: [2210.17323](https://arxiv.org/abs/2210.17323). URL: <https://arxiv.org/abs/2210.17323>.
- [16] J. Lin, J. Tang, H. Tang, S. Yang, X. Dang, C. Gan, and S. Han. “AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration”. In: *Proceedings of Machine Learning and Systems (MLSys)*. Best Paper Award. 2024. arXiv: [2306.00978](https://arxiv.org/abs/2306.00978). URL: <https://arxiv.org/abs/2306.00978>.
 - [17] G. Xiao, J. Lin, M. Seznec, H. Wu, J. Demouth, and S. Han. “SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models”. In: *Proceedings of the 40th International Conference on Machine Learning (ICML)*. 2023. arXiv: [2211.10438](https://arxiv.org/abs/2211.10438). URL: <https://arxiv.org/abs/2211.10438>.
 - [18] Z. Yao, R. Y. Aminabadi, M. Zhang, X. Wu, C. Li, and Y. He. “ZeroQuant: Efficient and Affordable Post-Training Quantization for Large-Scale Transformers”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2022.
 - [19] W. Shao, M. Chen, Z. Zhang, P. Xu, L. Zhao, Z. Li, K. Zhang, P. Gao, Y. Qiao, and P. Luo. “OmniQuant: Omnidirectionally Calibrated Quantization for Large Language Models”. In: *International Conference on Learning Representations (ICLR)*. Spotlight presentation. 2024. arXiv: [2308.13137](https://arxiv.org/abs/2308.13137). URL: <https://arxiv.org/abs/2308.13137>.
 - [20] S. Ma, H. Wang, L. Ma, L. Wang, W. Wang, S. Huang, L. Dong, R. Wang, J. Xue, and F. Wei. *The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits*. 2024. arXiv: [2402.17764](https://arxiv.org/abs/2402.17764). URL: <https://arxiv.org/abs/2402.17764>.
 - [21] V. Egiazarian, A. Panferov, D. Kuznedev, E. Frantar, A. Babenko, and D. Alistarh. *Extreme Compression of Large Language Models via Additive Quantization*. 2024. arXiv: [2401.06118](https://arxiv.org/abs/2401.06118). URL: <https://arxiv.org/abs/2401.06118>.
 - [22] Z. Liu, C. Zhao, I. Fedorov, B. Soran, D. Choudhary, R. Krishnamoorthi, V. Chandra, Y. Tian, and T. Blankevoort. “SpinQuant: LLM Quantization with Learned Rotations”. In: *International Conference on Learning Representations (ICLR)*. 2025. arXiv: [2405.16406](https://arxiv.org/abs/2405.16406). URL: <https://arxiv.org/abs/2405.16406>.
 - [23] Z. Liu, B. Oguz, C. Zhao, E. Chang, P. Stock, Y. Mehdad, Y. Shi, R. Krishnamoorthi, and V. Chandra. “LLM-QAT: Data-Free Quantization Aware Training for Large Language Models”. In: *Findings of the Association for Computational Linguistics: ACL 2024*. 2024. arXiv: [2305.17888](https://arxiv.org/abs/2305.17888). URL: <https://arxiv.org/abs/2305.17888>.
 - [24] Z. Liu et al. *Quantization Meets Reasoning: Exploring LLM Low-Bit Quantization Degradation for Mathematical Reasoning*. 2025. arXiv: [2501.03035](https://arxiv.org/abs/2501.03035). URL: <https://arxiv.org/abs/2501.03035>.
 - [25] G. Gerganov and llama.cpp contributors. *llama.cpp: Inference of Meta’s LLaMA model (and others) in pure C/C++*. Open-source project. 2023. URL: <https://github.com/ggml-org/llama.cpp>.
 - [26] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. “LoRA: Low-Rank Adaptation of Large Language Models”. In: *International Conference on Learning Representations (ICLR)*. 2022. arXiv: [2106.09685](https://arxiv.org/abs/2106.09685). URL: <https://arxiv.org/abs/2106.09685>.
 - [27] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer. “QLoRA: Efficient Finetuning of Quantized LLMs”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: [2305.14314](https://arxiv.org/abs/2305.14314). URL: <https://arxiv.org/abs/2305.14314>.
 - [28] S.-Y. Liu, C.-Y. Wang, H. Yin, P. Molchanov, Y.-C. F. Wang, K.-T. Cheng, and M.-H. Chen. “DoRA: Weight-Decomposed Low-Rank Adaptation”. In: *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Oral presentation. 2024. arXiv: [2402.09353](https://arxiv.org/abs/2402.09353). URL: <https://arxiv.org/abs/2402.09353>.
 - [29] C. Wu, Y. Gan, Y. Ge, Z. Lu, J. Wang, Y. Feng, Y. Shan, and P. Luo. “LLaMA Pro: Progressive LLaMA with Block Expansion”. In: *Proceedings of the 62nd Annual Meeting*

- of the Association for Computational Linguistics (ACL). 2024. arXiv: 2401.02415. URL: <https://arxiv.org/abs/2401.02415>.
- [30] A. Komatsuzaki, J. Puigcerver, J. Lee-Thorp, C. R. Ruiz, B. Mustafa, J. Ainslie, Y. Tay, M. Dehghani, and N. Hounsby. *Sparse Upcycling: Training Mixture-of-Experts from Dense Checkpoints*. 2022. arXiv: 2212.05055. URL: <https://arxiv.org/abs/2212.05055>.
 - [31] I. Beltagy, K. Lo, and A. Cohan. “SciBERT: A Pretrained Language Model for Scientific Text”. In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*. 2019. arXiv: 1903.10676. URL: <https://arxiv.org/abs/1903.10676>.
 - [32] S. Chithrananda, G. Grand, and B. Ramsundar. *ChemBERTa: Large-Scale Self-Supervised Pretraining for Molecular Property Prediction*. 2020. arXiv: 2010.09885. URL: <https://arxiv.org/abs/2010.09885>.
 - [33] R. Taylor, M. Kardas, G. Cucurull, T. Scialom, A. Hartshorn, E. Saravia, A. Poulton, V. Kerkez, and R. Stojnic. *Galactica: A Large Language Model for Science*. 2022. arXiv: 2211.09085. URL: <https://arxiv.org/abs/2211.09085>.
 - [34] K. Singhal, T. Tu, J. Gottweis, R. Sayres, E. Wulczyn, L. Hou, K. Clark, S. Pfohl, H. Cole-Lewis, D. Neal, et al. “Toward Expert-Level Medical Question Answering with Large Language Models”. In: *Nature Medicine* (2025). arXiv: 2305.09617. URL: <https://arxiv.org/abs/2305.09617>.
 - [35] C. Wu, X. Zhang, Y. Zhang, Y. Wang, and W. Xie. *PMC-LLaMA: Towards Building Open-source Language Models for Medicine*. 2023. arXiv: 2304.14454. URL: <https://arxiv.org/abs/2304.14454>.
 - [36] S. Gururangan, A. Marasović, S. Swayamdipta, K. Lo, I. Beltagy, D. Downey, and N. A. Smith. “Don’t Stop Pretraining: Adapt Language Models to Domains and Tasks”. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2020. URL: <https://aclanthology.org/2020.acl-main.740/>.
 - [37] Y. Luo, Z. Yang, F. Meng, Y. Li, J. Zhou, and Y. Zhang. *An Empirical Study of Catastrophic Forgetting in Large Language Models During Continual Fine-tuning*. 2023. arXiv: 2308.08747. URL: <https://arxiv.org/abs/2308.08747>.
 - [38] T. Schick, J. Dwivedi-Yu, R. Dessì, R. Raileanu, M. Lomeli, L. Zettlemoyer, N. Cancedda, and T. Scialom. “Toolformer: Language Models Can Teach Themselves to Use Tools”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. arXiv: 2302.04761. URL: <https://arxiv.org/abs/2302.04761>.
 - [39] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao. “ReAct: Synergizing Reasoning and Acting in Language Models”. In: *International Conference on Learning Representations (ICLR)*. 2023. arXiv: 2210.03629. URL: <https://arxiv.org/abs/2210.03629>.
 - [40] Anthropic. *Model Context Protocol Specification*. Open standard released November 2024. 2024. URL: <https://modelcontextprotocol.io/specification/>.
 - [41] P. Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2020. arXiv: 2005.11401. URL: <https://arxiv.org/abs/2005.11401>.
 - [42] Y. Ma et al. *SciAgent: Tool-augmented Language Models for Scientific Reasoning*. 2024. arXiv: 2402.11451. URL: <https://arxiv.org/abs/2402.11451>.
 - [43] J. Abramson, J. Adler, J. Dunger, R. Evans, T. Green, A. Pritzel, O. Ronneberger, L. Willmore, A. J. Ballard, J. Bambrick, et al. “Accurate Structure Prediction of Biomolecular Interactions with AlphaFold 3”. In: *Nature* 630 (2024), pp. 493–500. URL: <https://www.nature.com/articles/s41586-024-07487-w>.
 - [44] T. H. Trinh, Y. Wu, Q. V. Le, H. He, and T. Luong. “Solving Olympiad Geometry Without Human Demonstrations”. In: *Nature* 625 (2024), pp. 476–482. URL: <https://www.nature.com/articles/s41586-023-06747-5>.

- [45] Y. Zhang, X. Chen, B. Jin, S. Wang, S. Ji, W. Wang, and J. Han. “A Comprehensive Survey of Scientific Large Language Models and Their Applications in Scientific Discovery”. In: *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2024. arXiv: [2406.10833](https://arxiv.org/abs/2406.10833). URL: <https://arxiv.org/abs/2406.10833>.
- [46] Z. Zhang et al. *LLM4SR: A Survey on Large Language Models for Scientific Research*. 2025. arXiv: [2501.04306](https://arxiv.org/abs/2501.04306). URL: <https://arxiv.org/abs/2501.04306>.
- [47] A. Dubey et al. *The Llama 3 Herd of Models*. 2024. arXiv: [2407.21783](https://arxiv.org/abs/2407.21783). URL: <https://arxiv.org/abs/2407.21783>.
- [48] Gemma Team. *Gemma 3 Technical Report*. 2025. arXiv: [2503.19786](https://arxiv.org/abs/2503.19786). URL: <https://arxiv.org/abs/2503.19786>.
- [49] A. Yang, B. Yang, B. Hui, B. Zheng, B. Yu, C. Zhou, C. Li, C. Li, D. Liu, F. Huang, et al. *Qwen2.5 Technical Report*. 2024. arXiv: [2412.15115](https://arxiv.org/abs/2412.15115). URL: <https://arxiv.org/abs/2412.15115>.
- [50] M. Abdin, J. Aneja, H. Awadalla, A. Awadallah, A. A. Awan, N. Bach, A. Bahree, A. Bakhtiari, H. Behl, A. Benhaim, et al. *Phi-3 Technical Report: A Highly Capable Language Model Locally on Your Phone*. 2024. arXiv: [2404.14219](https://arxiv.org/abs/2404.14219). URL: <https://arxiv.org/abs/2404.14219>.
- [51] A. Q. Jiang, A. Sablayrolles, A. Roux, A. Mensch, B. Savary, C. Bamford, D. S. Chaplot, D. de las Casas, E. B. Hanna, F. Bressand, et al. *Mixtral of Experts*. 2024. arXiv: [2401.04088](https://arxiv.org/abs/2401.04088). URL: <https://arxiv.org/abs/2401.04088>.
- [52] DeepSeek-AI. *DeepSeek-V3 Technical Report*. 2024. arXiv: [2412.19437](https://arxiv.org/abs/2412.19437). URL: <https://arxiv.org/abs/2412.19437>.
- [53] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, and S. R. Bowman. *GPQA: A Graduate-Level Google-Proof Q&A Benchmark*. 2023. arXiv: [2311.12022](https://arxiv.org/abs/2311.12022). URL: <https://arxiv.org/abs/2311.12022>.
- [54] Y. Wang, X. Ma, G. Zhang, Y. Ni, A. Chandra, S. Guo, W. Ren, A. Arulraj, X. He, Z. Jiang, et al. “MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark”. In: *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. 2024. arXiv: [2406.01574](https://arxiv.org/abs/2406.01574). URL: <https://arxiv.org/abs/2406.01574>.
- [55] X. Sun, J. Lin, B. Bie, and B. Liu. *Neural Network Activation Quantization with Bitwise Information Bottlenecks*. 2020. arXiv: [2006.05210](https://arxiv.org/abs/2006.05210). URL: <https://arxiv.org/abs/2006.05210>.